Problem 1

$$I = \int_0^1 \int_0^x \int_x^1 e^{z^3}\, dz\, dy\, dx.$$

1. Sketch the solid E of integration.
2. Rewrite I in the order $dy\, dx\, dz$.
3. Evaluate I using the order in part (2).

Problem 2

Let E be the solid bounded by

$$z = x^2 + y^2 \qquad \text{and} \qquad z = 4.$$

1. Give cylindrical-coordinate bounds for E.
2. Use cylindrical coordinates to find the volume of E.
3. Evaluate $J = \iiint_E (x^2 + y^2)\, dV$ in cylindrical coordinates.

Problem 3

Let E be the solid above the cone and inside the sphere:

$$z = \sqrt{x^2 + y^2}, \qquad x^2 + y^2 + z^2 = 2z.$$

1. Give spherical-coordinate bounds for E.
2. Use spherical coordinates to find the volume of E.
3. Evaluate $J = \iiint_E (x^2 + y^2 + z^2)\, dV$ in spherical coordinates.